

VEDANT PANCHAL

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EDUCATION

Indian Institute of Technology, Roorkee

2018 - 2022 | Uttarakhand, India

Bachelor of Technology, Mechanical Engineering

WORK EXPERIENCE

Founding Engineer, Meritic - An AI Data Analyst product

Mar 2025 - Present | Bengaluru

- Developed modular backend for the **Darwin Agent Framework** — a multi-agent orchestrator, hierarchical agent-state store, and configuration system, with unified LLM-provider abstractions, agentic and non-agentic tools, and event-driven SSE streaming with selective emission of provider tokens, agent life-cycle events, thinking events, and tool outputs.
- Built a **tracing and analytics dashboard** enriching each LangGraph run with metadata — improving debugging efficiency, helping reduce LLM token costs to **10%-30%**, and helping rapid-prototype tools like Charting and Pandas Data-Analysis.
- Tech Stack: React/TS, Zod, Python, asyncio, FastAPI, Pandas, Langsmith, LangGraph, Redis, PostgreSQL, Docker, AWS.

IC, GenAI Client, Dubai

Dec 2023 - Present | Remote

- Hybrid Orchestrator: Developed a unique composite of LLM-based and menu-based bots with minimized hallucinations. Customers could deploy this on their own infrastructure. Optimized latency using concurrency and inference-streaming.
- Apart from support for open-source LLMs, RAG agents, conversation abilities, Orchestrator showed excellent adherence to upsert-ed SOPs and assigned bot persona. Modularized to use as a library with text or voice channels over REST APIs.
- Forked vocode library for custom use to achieve <2.5s perceived latency for bot-conversation over cellular calls. Used for high Customer Care traffic to assist clients' dept. Latency logs added for each of the STT, TTS, and telephony components.
- Effective collaboration with the business for product designing and requirement drafting. As the sole contributor to the engine of the technology, flew to Dubai Police HQ to partake in demo, discussions, and winning a contract for the product.
- Tech Stack: Event-driven Python, FastAPI, Supabase, Kong, Docker, Redis, LangChain, LlamaIndex, Langfuse, LiveKit

Software Developer, MathWorks

Jun 2022 - Dec 2023 | Bengaluru

- MATLAB Coder compiler: Added support to universally compile into High Level Synthesis System C compliant code. This was a doorway to roll-out features for widely compiler in the chip design industry. Tech Stack: MATLAB and C++.
- Fabricated a CI pipeline and a custom Linux environment to emulate hardware. This environment could be used in the CI services we offer to clients for unit-tests and system-tests for their automobile AUTOSAR applications. This feature is leveraged by customers for workflow across several hardware types. Tech Stack: AWS CDK and EKS, Linux, and Yocto.
- Developed and managed services in Go that processed GBs of customer artifacts to generate test-reports. Maintained a service mesh of these micro-services to simplify their communication interface. Used by customers that integrate MATLAB with other CI tools for automated testing. Tech Stack: Go, GraphQL, K8s, Terraform, Prometheus, and Istio.
- Upgraded CMake build system to use binary libraries from external projects on Git to accelerate the open-source C++ project for MATLAB-Apache Arrow interoperability. Tech Stack: CMake and Linux-tools for working with 'so' like 'ldd'
- Packaged a plugin to generate server and client SDK from an OpenAPI spec. Added support to generate docs. This effort helped to later adopt design-led development throughout many teams. Tech Stack: Maven, Go, and OpenAPI.

Founding Engineer, Explorin.io, Chandigarh

Oct 2020 - Apr 2022 | Remote

- Developed a platform for time-bound tasks, admin-configurable curriculum, and assignment checker. Tracked various aspects of a user's learning to provide well-detailed leader-board and progress analytics. Incorporated a pipeline to process platform metrics and analytics. Tech Stack: AWS, Django REST, Redis, Celery, ReactJS, Bootstrap, and jQuery.
- Developed a platform where students can deploy their Python or NodeJS applications. They can boot up instances of deployed applications in dev or prod environments with relevant configurations and create database users and tables for the same. Tech Stack: AWS, Django REST, NextJS, and Apache server.

SDE Intern, UniAcco, Mumbai

Aug 2020 - Sep 2020 | Remote

- Designed and developed a web-scraping pipeline and a notification delivery system for database changes.
- Delivered a PoC AWS Connect customer support system through cellular calls. Integrated webhooks for customer status updates using middleware in the business application. Tech Stack: SQL, Django, bs4, Selenium, AWS Connect

ACHIEVEMENTS

- Ranked in top 1.5 percentile in JEE-Advanced 2018.
- Ranked in top 2 percentile in UG IAPT Physics Olympiads.
- Qualified PoC (Blockchain based data-storage with RDF Schema) in Ethereum 3.0 Challenge.